

Information Management

Assignment 3

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Q2. Write SQL codes to get a list of: i. Students IDs (hint: from the takes relation) ii. Instructors iii. Departments

Online SQL interpreter

Run queries directly from the text box below; the university database schema and sample data have been preloaded. (Note: page may take a few seconds to load initially.). All query processing is done right in your browser using the SQLite database. Save the database and load it later, if you want your data to persist when you close the browser tab.
Click here [for tips on using SQLite](#) including SQL syntax variations.

Enter SQL commands here

```
1 SELECT DISTINCT ID FROM takes;
```

Execute

Save the db

Load an SQLite database file: No file chosen

ID
00128
12345
19991
23121
44553
45678
54321
55739
76543
76653
98765
98988

Enter SQL commands here

```
1 SELECT name FROM instructor;
```

Execute

Save the db

Load an SQLite database file: No file chosen

name
Srinivasan
Wu
Mozart
Einstein
El Said
Gold
Katz
Califieri
Singh
Crick
Brandt
Kim

Enter SQL commands here

```
1 SELECT dept_name FROM department;
```

Execute

Save the db

Load an SQLite database file: No file chosen

dept_name
Biology
Comp. Sci.
Elec. Eng.
Finance
History
Music
Physics

Q. 3.

i. Find the ID and name of each student who has taken at least one Comp. Sci. course; make sure there are no duplicate names in the result.

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Enter SQL commands here

```
1 SELECT DISTINCT student.ID, student.name
2 FROM student
3 JOIN takes ON student.ID = takes.ID
4 JOIN course ON takes.course_id = course.course_id
5 WHERE course.dept_name = 'Comp. Sci.';
```

Execute

Save the db

Load an SQLite database file:

No

ID	name
00128	Zhang
12345	Shankar
45678	Levy
54321	Williams
76543	Brown
98765	Bourikas

Q3. ii Add grades to the list

Enter SQL commands here

```
1 SELECT DISTINCT student.ID, student.name, takes.grade
2 FROM student
3 JOIN takes ON student.ID = takes.ID
4 JOIN course ON takes.course_id = course.course_id
5 WHERE course.dept_name = 'Comp. Sci.';
```

Execute

Save the db

Load an SQLite database file:

No

ID	name	grade
00128	Zhang	A
00128	Zhang	A-
12345	Shankar	C
12345	Shankar	A
45678	Levy	F
45678	Levy	B+
45678	Levy	B
54321	Williams	A-
54321	Williams	B+
76543	Brown	A
98765	Bourikas	C-
98765	Bourikas	B

Q. 3.

iii. Find the ID and name of each student who has not taken any course offered before 2017

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Enter SQL commands here

```
1 SELECT ID, name
2 FROM student
3 WHERE ID NOT IN (
4     SELECT ID
5     FROM takes
6     WHERE year < 2017
7 );
```

Execute

Save the db

Load an SQLite database file:

Choose File

ID	name
00128	Zhang
12345	Shankar
19991	Brandt
23121	Chavez
44553	Peltier
45678	Levy
54321	Williams
55739	Sanchez
70557	Snow
76543	Brown
76653	Aoi
98765	Bourikas
98988	Tanaka

Q. 3.

iv. For each department, find the maximum salary of instructors in that department. You may assume that every department has at least one instructor.

Enter SQL commands here

```
1 SELECT dept_name, MAX(salary) AS max_salary
2 FROM instructor
3 GROUP BY dept_name;
```

Execute

Save the db

Load an SQLite database file:

Choose File

No

dept_name	max_salary
Biology	72000
Comp. Sci.	92000
Elec. Eng.	80000
Finance	90000
History	62000
Music	40000
Physics	95000

Q. 3.

v Find the lowest, across all departments, of the per-department maximum salary computed by the preceding query

Enter SQL commands here

```
1 SELECT MIN(max_salary)
2 FROM (
3     SELECT MAX(salary) AS max_salary
4     FROM instructor
5     GROUP BY dept_name
6 ) AS dept_maxes;
```

Execute

Save the db

Load an SQLite database file:

MIN(max_salary)
40000

Q. 3.

vi. Add names to the list

Enter SQL commands here

```
1 SELECT MIN(max_salary)
2 FROM (
3     SELECT MAX(salary) AS max_salary
4     FROM instructor
5     GROUP BY dept_name
6 ) AS dept_maxes;
```

Execute

Save the db

Load an SQLite database file:

MIN(max_salary)
40000

Q.5 Write SQL query to find the number of students in each section. The result columns should appear in the order “courseid, secid, year, semester, num”. You do not need to output sections with 0 students.

Enter SQL commands here

```
1 SELECT course_id, sec_id, year, semester, COUNT(ID) AS num
2 FROM takes
3 GROUP BY course_id, sec_id, year, semester;
```

Execute

Save the db

Load an SQLite database file:

No file ch

course_id	sec_id	year	semester	num
BIO-101	1	2017	Summer	1
BIO-301	1	2018	Summer	1
CS-101	1	2017	Fall	6
CS-101	1	2018	Spring	1
CS-190	2	2017	Spring	2
CS-315	1	2018	Spring	2
CS-319	1	2018	Spring	1
CS-319	2	2018	Spring	1
CS-347	1	2017	Fall	2
EE-181	1	2017	Spring	1
FIN-201	1	2018	Spring	1
HIS-351	1	2018	Spring	1
MU-199	1	2018	Spring	1
PHY-101	1	2017	Fall	1